Faculty of Computer Applications and Information Technology
M. Sc. (IT) – Sem 3
Network Analysis Tools
CEC Assignment 1

Note: Kindly submit your assignment till 01st September 2025 on the below drive link

https://drive.google.com/drive/folders/1SzgY0AYyXROE_S7zIfCIkTBqa8YrTKgG?usp=sharing

Please prepare a PDF file of your assignment, save it with your enrollment number as the file name, and upload it to the drive link shared above.

Open the file wiresharksamplecapture.pcapng in Wireshark.

1. How many total packets are present?
2. List the top 3 protocols by frequency.
3. Which IP address appears most frequently as a source?
4. Apply the filter ip.addr == <most frequent IP> and count the number of packets.
5. Apply the filter tcp – how many TCP packets are there?
6. Identify the source and destination MAC addresses of the first packet.
7. Apply a filter based on the source MAC address.
8. Find a complete TCP conversation (use Follow TCP Stream).
9. Create a filter to show only TCP SYN packets: tcp.flags.syn == 1 and tcp.flags.ack == 0. How many packets match this filter?
10. Add a column for “Time delta from previous captured packet”.
11. Identify the largest gap in packet arrival.
12. What is the length of the smallest and largest packet?

13. Create a color rule to highlight all TCP SYN packets in green.
14. Create a filter button for HTTP packets and assign a custom label.
15. Show only DNS queries not responses.(Hint: dns.flags.response == 0)
16. Show TCP packets where the destination port is 443. (Hint:tcp.dstport == 443)
17. Display packets larger than 500 bytes. (Hint: frame.len > 500)
18. Show all ICMP Echo Request packets. (Hint: icmp.type == 8)
19. Show traffic where the source IP is 145.254.160.237 and the destination port is 80.
20. Show traffic from either 145.254.160.237 or 172.16.18.160
21. Exclude ARP packets from display
22. Find the number of HTTP GET requests.
23. Filter packets involved in a TCP handshake. (Hint: tcp.flags.syn == 1 || tcp.flags.ack == 1)
24. Check if there was any retransmission. (Hint: tcp.analysis.retransmission)
25. Show only packets captured between 10 and 20 seconds. (Hint: frame.time_relative >= 10 && frame.time_relative <= 20)
26. Show HTTP requests where Host contains testfire.net (Hint: http.host matches "testfire.net")
27. Show DNS queries ending with .edu. (Hint: dns.qry.name matches ".edu$")
28. Extract first 3 bytes of destination MAC. (Hint: eth.dst[0:3])
29. Mark all packets with TCP RST. (Hint: tcp.flags.reset == 1)
30. Show all traffic except TCP & UDP.(Hint: !(tcp || udp))
31. Display packets with TCP destination port in {80, 443, 8080}.(Hint: tcp.dstport in {80 443 8080})

32. Show packets where the source IP is either 145.254.160.237, 172.16.18.160, or 65.208.228.223.(Hint: ip.src in {145.254.160.237, 172.16.18.160, 65.208.228.223.})

33. Display packets where the first byte of the source IP is 192. (Hint: ip.src[0:1] == 192)

34. Extract the first 4 bytes of packet payload. (Hint: frame[0:4])

35. Show HTTP requests where the URI contains /login. (Hint: http.request.uri matches "login")

36. Show only ICMP packets where the code is not 0. (Hint: icmp.code != 0)

37. Find HTTP requests that contain the word 'login'.

38. Find a ping request and reply and Write the Round-Trip Time (RTT).

39. Use File → Export Objects → HTTP.
 Export one object (image/text file).
 Note the file name and size.

40. Apply filter `http`.
 Follow one HTTP stream.
 Write a short note on the data exchanged.